Fiscal Year:	FY 2013	**Task Last Updated:**	FY 06/13/2013
PI Name:	Tash, Joseph S Ph.D.		
Project Title:	Long Term Space Flight Impacts on Male Reproductive Health		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology (2) Developmental Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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City:	Kansas City	**State:**	KS
Zip Code:	66160-0001	**Congressional District:**	3
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2007 Space Flight: Bion-M1 NNH07ZTT001N
Start Date:	08/01/2009	**End Date:**	07/31/2015
No. of Post Docs:	2	**No. of PhD Degrees:**	3
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Smith, Jeffrey	**Contact Phone:**	650-604-0880
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Flight Program:	Bion/Shuttle		
Flight Assignment:	STS-131 ; STS-133 ; STS-135 ; Bion M1 NOTE: Extended to 7/31/2015 per A. Chu/ARC and NSSC information (Ed., 8/5/14) NOTE: Extended to 7/31/2014 per A. Chu/ARC (Ed., 5/7/13)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Roby, Katherine (University of Kansas Medical Center) Gupta, Vijayalaxmi (University of Kansas Medical Center) Holets, Lesya (University of Kansas Medical Center)		
Grant/Contract No.:	NNX09AP04G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Published data demonstrates that short term microgravity (μG) (~2 wks) produces small but significant reductions in spermatogenesis in rats. However, long term ground-based μG models using the rodent hindlimb unloading (HLS) (> 3 wks), with surgical prevention of cryptorchidism, show dramatic testicular atrophy and loss of spermatogenesis leading to sterility. These results and related literature in humans suggest that longer duration space flight may have negative impacts on spermatogenesis and compromise male health and increase cancer risk. Thus, the testable hypothesis driving this proposal is: Exposure to μG 3 wks or longer causes disruption of spermatogenesis in mice. This proposal addresses the Bioastronautic Roadmap (BR) Risk 30: Chronic and Degenerative Tissue Risks in that long term μG-induced testicular atrophy will increase the risk for long term health problems post-flight including increased cancer risk and sterility. The hypothesis will be tested with the following specific aims: 1. Identify specific histopathology and dysfunction in spermatogenesis and testis function in mice exposed to LEO for 30d (as determined by mission duration) and compare to ground-based controls. 2. Quantify declines in sperm motility and morphology in cauda epididymal sperm from male mice exposed to LEO for 30d (as determined by mission duration) and compare to ground-based controls. 3. Determine dysregulation in gene transcription of markers of spermatogenic regulation and inflammatory markers in testes from mice exposed to LEO for 30d (as determined by mission duration) and compare to ground-based controls. 4. Determine declines in fertility and fecundity of male mice exposed to μG (This aim will depend upon availability of flight animals for mating trials). The opportunity to expose male mice for 30d of μG in the new BION flights represents the first to directly evaluate this reproductive health problem directly in long term μG under properly controlled conditions. These experiments will validate the ground-based long term HLS rodent model and provide proof that male reproductive health is put at risk by extended exposure to μG. The experiments will provide an initial data set for development of appropriate countermeasures to the negative impact of long term space flight on endocrine and reproductive physiology compromising male health.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Female reproductive studies: The observed down-regulation of estrogen receptor ERa, combined with the histological changes, suggests that the animals stopped cycling in space flight and estrogen support was severely impaired. The observations are also significant in that the results were verified on three separate Space Shuttle experiments (STS-131, STS-133, and STS-135). If ERa is also down regulated in other tissues, this would indicate a common mechanism underlying space-flight induced degradation of other estrogen dependent systems including bone loss, muscle weakness, immune response, and wound repair. A broad effect of space flight on ERa receptor levels and/or signaling will offer a common pathway to generate countermeasures for space flight. Since similar changes occur in the same organ systems during the aging process, the countermeasures would have application for improving quality of life in the same organ systems during the aging process. Male reproductive health: The experiments outlined in this proposal will be the first to determine directly, under controlled space flight conditions, whether exposure to long microgravity (μG) disrupts sperm production and compromises male reproductive health in mammals. The unique opportunities offered in the new BION 30d+ rodent flights are the first to extend space flight beyond all previous μG exposures for rodent models on any flight platform. As such, the planned experiments will be the first to directly assess a long term health risk that has been identified by ground-based space flight animal and human models. Thus the hypothesis driving the research: that long term space flight has negative impacts on male reproductive health. Specific Aim 1. Identify specific histopathology and dysfunction in spermatogenesis and testis function in mice exposed to LEO for 30d (as determined by mission duration) and compare to ground-based controls. Specific Aim 2. Quantify declines in sperm motility and morphology in cauda epididymal sperm from male mice exposed to LEO for 30d (as determined by mission duration) and compare to ground-based controls. Specific Aim 3. Determine dysregulation in gene transcription of markers of spermatogenic regulation and inflammatory markers in testes from mice exposed to LEO for 30d (as determined by mission duration) and compare to ground-based controls. Space flight impacts on reproductive health include both microgravity and radiation environmental factors. Space flight changes are an excellent accelerated analogue to the aging process on Earth. The 30 day BION M1 mission will combine the longest exposure and the largest number of mice in a single mission and represent a significant duration relative to the complete spermatogenesis cycle in the testis. Changes in sperm motility, spermatogenesis, gene transcription, and DNA damage in long term space flight will offer the first comparison for identical male reproductive health outcome measures for the normal aging process on Earth.
Task Progress:	BION M1 Male reproductive studies: The flight portion of the BION M1 mission was completed at the IMBP in Moscow RU in May, 2013. Testes and epididymides were received from 6 male mice, and from 8 vivarium control animals. Live sperm were processed from all the animals and sperm motility was recorded at the IMBP laboratory. The tissues and sperm motility recordings as well as live-dead stained sperm samples have successfully arrived back at Dr. Tash's laboratory. Both the flight and vivarium control tissue samples are now being processed for tissue histopathology. The samples for gene transcription changes are being stored at -80C until the ground control samples are obtained when we return to IMBP later this year. Initial analysis has shown that the individual animal variation of testis weights was remarkably consistent with a variance of only 0.0001 mg. This remarkable consistency within groups was similar for the vivarium controls. Although the sperm motility recordings remain to be quantitatively analyzed using our CASA system, initial visual assessments of the raw video files have shown that the sperm motility was very good in both the flight animals and the vivarium controls. The sperm motility recordings represent the first ever analysis of live sperm that completed spermatogenesis and spermiogenesis in microgravity.

Bibliography Type:	Description: (Last Updated: 01/24/2022)
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Abstracts for Journals and Proceedings	Holets LM, Gupta V, Roby KF, Tash JS. "Spaceflight inhibits ovarian follicle development, induces down regulation of estrogen receptor alpha, and alters metabolic pathways and gene expression in mouse uterus." 45th Annual Meeting of the Society for the Study of Reproduction, Pennsylvania State University, State College, PA, August 12-15, 2012. 45th Annual Meeting of the Society for the Study of Reproduction, Pennsylvania State University, State College, PA, August 12-15, 2012. http://www.ssr.org/Documents/2012-09-04ScheduleAndAbstracts.pdf ; accessed 6/14/13. , Aug-2012